QA COPILOT · ARCHITECTURE OVERVIEW

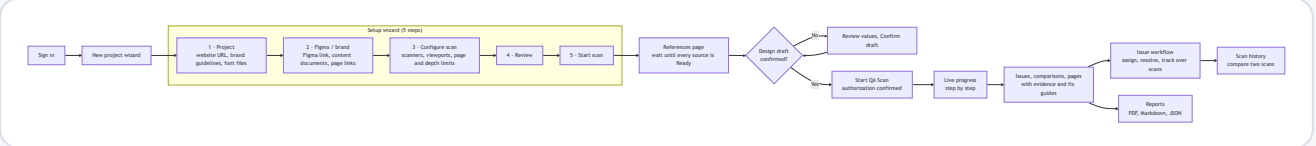
# How QA Copilot tests a website

QA Copilot crawls a website in a real browser, runs seven automated QA scanners, compares the pages with the project's Figma design, brand guidelines and approved content, and turns the evidence into grouped issues and reports. This page shows the workflow, the scan pipeline, the system architecture and the technology used.

TypeScript monorepo | 2 apps · 10 packages | Node.js 22 | PostgreSQL | Playwright · Chromium

## 1. What the tester does

The setup wizard collects the six inputs. Extracted design values stay a draft until someone confirms them; a scan uses the references exactly as they are when it starts.




### Six inputs

Website URL, brand guidelines, font files, Figma design URL, content documents and page links.

### Readiness guard

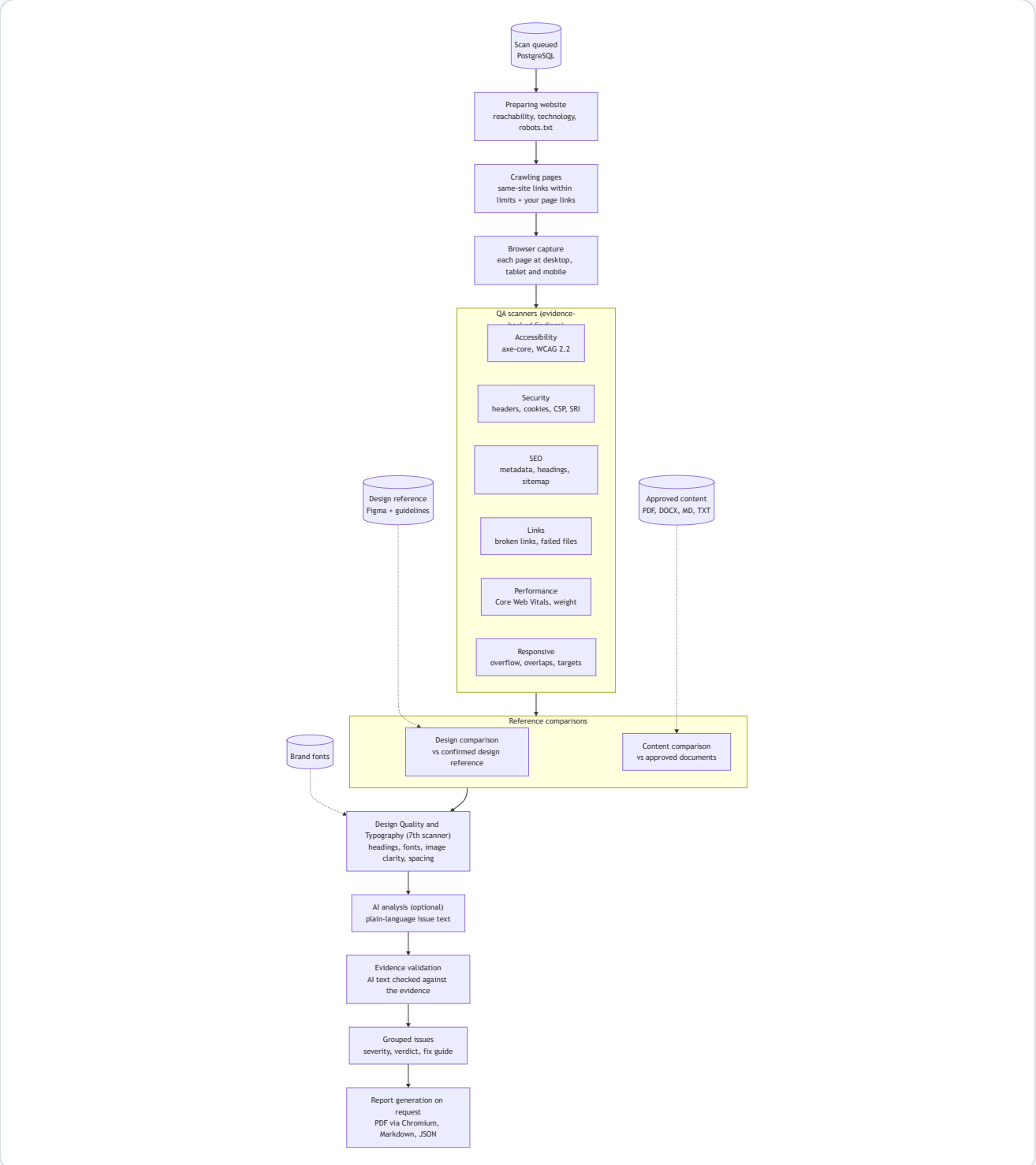
Start QA Scan stays disabled while the design draft is unconfirmed or sources are still processing, unless the tester ticks "Start anyway".

### Re-scans

Issues keep their identity across scans, so a fix shows as Resolved only when a later scan no longer finds it.

## 2. What happens during a scan

The worker runs every scan as a fixed sequence of steps. Each step records its status, reason and finding count, so a skipped or failed step is reported, never shown as completed.

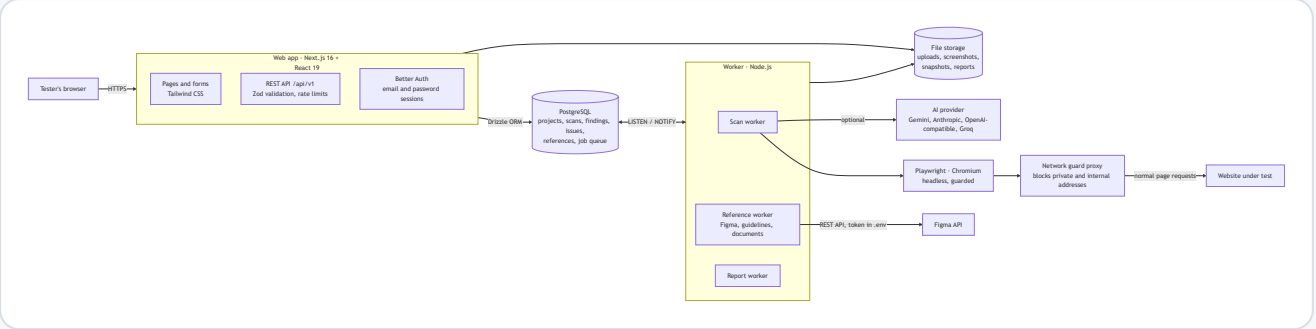



**Confirmed** finding: the evidence proves it. **Potential**: needs a person to check.

**Unable to verify**: listed with the reason, never guessed.

## 3. System architecture

Two processes share one PostgreSQL database. The web app accepts requests and queues work; the worker picks it up immediately through PostgreSQL LISTEN/NOTIFY and does all crawling, browser work and calls to Figma and the AI provider.




### Safe by default

Every outgoing request goes through SSRF rules: scans cannot reach private networks, and only normal page requests are made.

### Secrets stay server-side

Figma and AI keys live in `.env` or are encrypted per project; they are never sent to the browser.

### Quota aware

Figma reads check the file version first, and AI calls are paced to the provider's free-tier limits.

## 4. Tech stack

### Language and runtime — FOUNDATION

| | |
|---|---|
| **TypeScript** | All apps and packages, strict typing |
| **Node.js 22** | Runtime for web app and worker |
| **npm workspaces** | Monorepo with shared packages |
| **Zod 4** | Config, API and data validation |
| **Pino** | Structured logging |

### Web app — APPS/WEB

| | |
|---|---|
| **Next.js 16** | App Router, server components, API routes |
| **React 19** | User interface |
| **Tailwind CSS 4** | Styling |
| **Lucide** | Icons |
| **Better Auth** | Sign-up, sign-in, sessions |

### Data — PACKAGES/DB, STORAGE

| | |
|---|---|
| **PostgreSQL** | All data, plus the job queue (LISTEN/NOTIFY) |
| **Drizzle ORM** | Schema, queries, migrations |
| **node-postgres (pg)** | Database driver |
| **Local file storage** | Uploads, screenshots, DOM snapshots, reports |

### Crawling and browser — CRAWLER, BROWSER

| | |
|---|---|
| **Playwright + Chromium** | Page capture at three viewports, PDF reports |
| **undici** | HTTP client for crawling and link checks |
| **Cheerio** | HTML parsing |
| **robots-parser** | robots.txt rules |
| **fast-xml-parser** | Sitemaps |

## QA scanners

PACKAGES/SCANNERS

| | |
|---|---|
| **axe-core** | Accessibility (WCAG 2.2 A/AA) |
| **Chrome DevTools data** | Core Web Vitals, rendered fonts |
| **culori** | Color distance (ΔE) for design checks |
| **diff** | Word-level content comparison |
| **Own rule engines** | Security, SEO, links, responsive, design quality |

## References

PACKAGES/REFERENCE

| | |
|---|---|
| **Figma REST API** | Text and color styles, buttons, containers |
| **unpdf** | PDF content documents |
| **mammoth** | Word (DOCX) documents |
| **mdast** | Markdown documents and guidelines |
| **Font validator** | TTF, OTF, WOFF, WOFF2 brand fonts |

## AI

PACKAGES/AI

| | |
|---|---|
| **Google Gemini** | Current provider (free tier) |
| **Anthropic SDK** | Claude models, optional |
| **OpenAI-compatible, Groq** | Alternative providers |
| **Grounding checks** | AI text accepted only if the evidence supports it |

## Security

PACKAGES/SECURITY

| | |
|---|---|
| **ipaddr.js** | Private-address (SSRF) blocking |
| **Guard proxy** | Filters every browser request |
| **Upload checks** | File type, size and content validation |
| **Encryption** | Per-project secrets encrypted at rest |

## Quality and delivery

TOOLING

| | |
|---|---|
| **Vitest** | Unit and integration tests |
| **Playwright Test** | End-to-end tests with axe audits |
| **ESLint, Prettier** | Linting and formatting |
| **Docker Compose** | PostgreSQL, web app and worker deployment |

## 5. Code map

| FOLDER | RESPONSIBILITY | MAIN TECHNOLOGY |
|---|---|---|
| apps/web | User interface, REST API, authentication, uploads | Next.js, React, Better Auth, Tailwind |
| apps/worker | Scan pipeline, reference processing, report generation, storage cleanup | Node.js, Playwright |
| packages/core | Shared types, configuration, scanner list, explanations and fix guides | TypeScript, Zod, Pino |
| packages/db | Database schema, migrations, repositories | Drizzle ORM, PostgreSQL |
| packages/crawler | Guarded HTTP fetching, link extraction, robots.txt, crawl | undici, Cheerio |
| packages/browser | Browser launch, page capture, in-page measurement scripts, PDF | Playwright, Chromium |
| packages/scanners | Seven QA scanners and the design and content comparisons | axe-core, culori, diff |
| packages/reference | Figma reading, guideline extraction, document parsing | Figma API, unpdf, mammoth |
| packages/ai | Issue grouping, AI interpretation, grounding | Gemini, Anthropic SDK |
| packages/reports | Report content and layout (HTML, Markdown, JSON) | TypeScript |
| packages/security | SSRF rules, upload and font validation, encryption | ipaddr.js |
| packages/storage | File storage for uploads, screenshots and reports | Node.js file system |

Taken from the QA Copilot repository on 23 October 2026 (package manifests, pipeline code and configuration). Versions: Next.js 16.3, React 19.3, Playwright 1.63, Drizzle 0.45, Zod 4.6, Node.js 22.